

GlowBeauty Premium Skincare — Audience Feed Specification

One-page data specification · Simulated brand request · Reference date 2026-09-21 (90-day window) · All data synthetic

Feed definition

Area	Definition
Business objective	Reach high-intent skincare shoppers for a premium skincare campaign
Target population	Customers with skincare engagement in the last 90 days
Exclusions	Opted-out users · missing/invalid email identifiers · inactive records
Required identifiers	Customer ID; hashed email (SHA-256) preferred for matching
Required fields	Hashed email, ZIP code, product category, last activity date, engagement score
Delivery format	CSV, one row per customer, UTF-8
Delivery cadence	Daily
Delivery method	S3 (mock: s3://glowbeauty-feeds/audience/) or SFTP drop — PGP-encrypted in production
Quality thresholds	≥ 95% required-field completeness · ≥ 70% email/hash matchability
Validation checks	Schema · duplicates · nulls · invalid emails · coverage · opt-out leakage
Known limitations	Missing identifiers reduce addressable audience and match rate

Field definitions

Column	Type	Description
customer_id	string	Deduplicated customer identifier (survivorship rule applied)
email	string	Valid-format email; rows without one are excluded
hashed_email	string	SHA-256 hex of lowercase email; missing values kept but flagged
zip_code	string	5-digit ZIP; missing values kept and reported
product_category	string	'Skincare' (feed scope)
last_activity_date	date	Most recent in-window skincare event
total_purchases	int	In-window purchases (valid products only)
total_spend	decimal	In-window spend (valid products only)
engagement_score	int	Weighted intent: page_view=1, product_view=2, add_to_cart=5, purchase=10 (cap 100)
premium_interaction_flag	int	1 if any premium skincare interaction in window
engagement_tier	string	high (≥10) / medium (4–9) / low (<4), calibrated to window

Selection logic (summary)

- Join **customers** → **events** → **products** → **orders**; keep skincare events in the last 90 days (future-dated and orphaned records dropped).
- Deduplicate** customers by customer_id (survivorship: usable email → hashed email → most recent).
- Exclude** opted-out customers and rows with missing/invalid emails.
- Aggregate per customer: last activity, event counts, engagement score, premium flag, purchases, spend.
- Keep** rows missing hashed_email (matchability input, not a delivery blocker) — the tradeoff analyzed in the findings report.

Validation: `sql/03_data_quality_validation.sql` · automated check: `python/feed_validation.py`.